Materials Science C - Materials Science - Monta Vista Science Olympiad (Div. C) - 11-14-2025

Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GCM)

Instructions (shown before students start the test)

good luck & show work for all math!

Introduction (shown after students start the test)

1. (3.00 pts) Which 4th century Roman artifact is an example of dichoric glass?

(Mark **ALL** correct answers)

- A) Mask of Agamemnon
- B) Lycurgus Cup
- C) Portland Vase
- D) Standard of Ur
- E) Venus of Willendorf
- F) Bust of Pericles

2. (3.00 pts) Who was the professor who coined the term "nanotechnology" and when?

(Mark **ALL** correct answers)

- A) Richard Feynman, 1959
- B) Donald Eigler, 1989
- C) Sumio Iijima, 1991
- D) Norio Taniguchi, 1974
- E) Paul Alivisatos, 1996
- F) K. Eric Drexler, 1968

3. (3.00 pts) What best describes the mechanism of photoablation?

- A) Using the photoelectric effect to release electrons from a material surface
- B) Chemical etching of surface atoms using reactive gases under UV exposure
- C) Using a photosensitive material under light for selective removal to create a pattern
- D) Using UV energy to break molecular bonds and remove material from a surface

4. (3.00 pts) Which of the following is not a carbon based nanomaterial?

(Mark **ALL** correct answers)

- A) Graphene
- B) Fullerene
- C) Carbon dots
- D) Graphite
- E) Diamond

- [ ] F) Pyramid of Khufu

5. (3.00 pts) Which of the following is a carbon nanotube structure?

(Mark **ALL** correct answers)

- [ ] A) Zig-zag
- [ ] B) Armchair
- [ ] C) Chrial

6. (3.00 pts) Which of the following types of microscopy allows for a resolution of exactly 10^{-10} to 10^{-7}m?

(Mark **ALL** correct answers)

- [ ] A) Optical
- [ ] B) SEM
- [ ] C) TEM
- [ ] D) AFM
- [ ] E) SPM
- [ ] F) SAM

7. (3.00 pts) Young's modulus is also known as:

(Mark **ALL** correct answers)

- [ ] A) Bulk Modulus
- [ ] B) Shear Modulus
- [ ] C) Elastic Modulus
- [ ] D) Flexural Modulus
- [ ] E) Hardness
- [ ] F) Toughness

8. (3.00 pts) What type of structure is a fullerene?

- ○ A) 0D
- ○ B) 1D
- ○ C) 2D
- ○ D) 3D

9. (3.00 pts) What type of structure is graphene?

- ○ A) 0D
- ○ B) 1D
- ○ C) 2D
- ○ D) 3D

10. (3.00 pts) What are nanotrusses?

(Mark **ALL** correct answers)

- A) Ceramic nanoparticles sintered under high pressure to form bulk nanostructured ceramics with improved toughness.
- B) Amorphous silica aerogels formed through sol–gel processing to achieve low thermal conductivity and low weight.
- C) Nanocrystalline ceramic coatings produced by plasma spraying for enhanced wear and oxidation resistance.
- D) Polycrystalline oxide fibers synthesized via electrospinning for reinforcement in ceramic matrix composites.
- E) Nanoscale ceramic lattices with truss-like geometry that achieve high stiffness at extremely low density.
- F) Porous ceramic membranes with interconnected nanopores used for selective molecular filtration in chemical processing.

11. (3.00 pts) Which of the following is not used in photolithography?

(Mark **ALL** correct answers)

- A) Photomask
- B) Photoresist
- C) Photoablation
- D) Substrate
- E) Photograph
- F) Pyramid of Khafre

12. (3.00 pts) Which compound is known for aiding in UV/IR protection?

(Mark **ALL** correct answers)

- A) Silicon dioxide (SiO_2)
- B) Titanium dioxide (TiO_2)
- C) Calcium carbonate ($CaCO_3$)
- D) Zinc sulfate ($ZnSO_4$)
- E) Aluminum oxide (Al_2O_3)
- F) Iron(III) oxide (Fe_2O_3)

13. (3.00 pts)
George Washington performs x-ray diffraction on a material and he observes that the peaks on his x-ray diffraction graph are very broad. What is the main reason for this?

- A) Large crystal size
- B) Small crystal size
- C) Longer wavelength
- D) Shorter wavelength

14. (3.00 pts) Which of the following is a 3D volume defect?

(Mark **ALL** correct answers)

- A) Precipitate
- B) Interphase Boundary
- C) Disclination
- D) Dispiration
- E) Faulted Region

- [ ] F) Stacking Faults

15. (3.00 pts)
Oh no! Julius Caesar is about to be stabbed by a group of senators, but he knows that they wont be able to use their daggers. This is because last night, Caeser put all of their weapons through this very cheap, quick, and cost-effective method to reduce the edge of the sharp blades into tiny metal nanoparticles. Which process did Caeser employ to save his life?

(Mark **ALL** correct answers)
- [ ] A) Chemical etching
- [ ] B) Physical vapor deposition
- [ ] C) Extrusion
- [ ] D) Ball milling
- [ ] E) Molecular beam epitaxy
- [ ] F) Electrospinning

16. (3.00 pts) A ray of light hits a diamond (n=2.42) at a 30º angle from the material surface. What angle is the light ray refracted at relative to the normal?

17. (5.00 pts) Ella Yao has a block of iron (BCC & radius = 1.26Å). What is her iron's unit cell length?

18. (5.00 pts) Ella decides to perform x-ray diffraction on her block. She uses a Cu Kα laser (λ = 1.54Å) in the (1,1,1) direction on her block. At what angle will the laser diffract?

19. (5.00 pts) Ella wants to find the density of her iron block, but she doesn't have a scale! Help Ella find the theoretical density of her iron block.

20. (5.00 pts) In a metal, the vacancy formation energy is q=1.5eV (note 1 eV = 1.602E-19 J). Calculate the concentration of vacancies (%) at T=1000K.

21. (3.00 pts) Which of the following best describes the Zener pinning effect?

(Mark **ALL** correct answers)
- A) The acceleration of grain boundary migration through enhanced boundary diffusion in the presence of fine precipitate atoms at high temperatures.
- B) The stabilization of dislocation substructures through dynamic processes that reduce stored strain energy without significant boundary migration.
- C) The alignment and elongation of second-phase particles along the direction of applied stress during plastic deformation.
- D) The preferential nucleation of recrystallized grains on incoherent second-phase particles during annealing, promoting heterogeneous grain refinement.
- E) The inhibition of grain boundary migration in a polycrystalline material caused by the interaction between fine particles or crystals.
- F) Who's Zener?

22. (3.00 pts) What compounds are often used in Chemical Vapor Deposition (CVD)?

(Mark **ALL** correct answers)
- A) metal hydrides
- B) halides
- C) carbonyls
- D) alkyls

23. (3.00 pts) Which of the following correctly matches the purpose of the following nanomaterials, often found in fabric, in that order: Nanoscale silver, silica, and zinc oxide?

(Mark **ALL** correct answers)
- A) Antimicrobial protection, stain resistance, and UV protection
- B) UV protection, waterproofing, and antimicrobial protection
- C) Antimicrobial protection, waterproofing, and UV protection
- D) Waterproofing, antimicrobial protection, and stain resistance
- E) UV protection, self-cleaning, and antimicrobial protection
- F) Quartz

24. (3.00 pts) For whom is buckministerfullerene named after?

(Mark **ALL** correct answers)
- A) R. Buckminster Fuller
- B) B. Buckminster Fuller
- C) Richard Buckman
- D) John Fullerene
- E) Buckminster Fullerene
- F) William Backflip

25. (3.00 pts) How many carbon atoms are present in the structure of buckministerfullerene?

(Mark **ALL** correct answers)
- A) 32
- B) 48
- C) 60

- D) 72
- E) 84
- F) 1

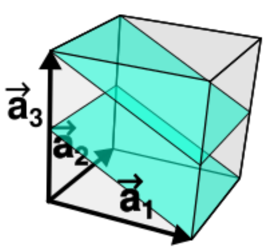
26. (3.00 pts)
Ella, who is extremely interested in nanomaterials and nanotechnology, decides to meet with the one and only Mr. Fuller to discuss her interests. Mr. Fuller is fed up about her ranting on about how she uses fullerenes in her everyday life and asks her to instead talk about graphene. Ella then remembers that she has a friend named Young who has a modulus for graphene. Ella talks about this and mentions that Young's modulus for graphene is:

(Mark **ALL** correct answers)
- A) 1 TPa
- B) 1 GPa
- C) 1 EPa
- D) 1 MPa
- E) 1 Pa
- F) 1 atm

27. (3.00 pts)
Unfortunately, Mr. Fuller is not amused, and in a cruel twist of fate, mentions his friend named Flexural. It turns out that Flexural also has a modulus, although it is quite different fro Young's modulus, which "is a mechanical property of solid materials that measures the tensile or compressive stiffness when the force is applied lengthwise." What is Flexural modulus?

(Mark **ALL** correct answers)
- A) The tendency for a material to resist compression.
- B) The tendency for a material to resist fracturing.
- C) The tendency for a material to resist twisting.
- D) Tendency for a material to resist bending.
- E) The tendency for a material to resist stretching.
- F) The tendency for a material to resist the force of gravity.

28. (3.00 pts)



Which miller indices plane shown here?

(Mark **ALL** correct answers)
- A) 100
- B) 101
- C) 102
- D) 002
- E) 010
- F) 020

29. (3.00 pts) Which of the following best defines the pauli exclusion principle?

(Mark **ALL** correct answers)

- ☐ A) No two electrons in an atom can have the same set of four quantum numbers.
- ☐ B) No two electrons in an atom can have the same set of three quantum numbers.
- ☐ C) No two electrons in an atom can have the same set of two quantum numbers.
- ☐ D) No three electrons in an atom can have the same set of four quantum numbers.
- ☐ E) No three electrons in an atom can have the same set of two quantum numbers.
- ☐ F) No electrons in an atom can have the same set of quantum numbers.

30. (3.00 pts) At what rate does typical molecular beam epitaxy (MBE) occur at?

(Mark **ALL** correct answers)

- ☐ A) 0.01–1 nanometer per second
- ☐ B) 1–10 micrometers per second
- ☐ C) 10–100 micrometers per minute
- ☐ D) 1–10 millimeters per hour
- ☐ E) 0.1–1 micrometer per hour
- ☐ F) 1 meter per minute

31. (3.00 pts) How many dimensions of confinement is exhibited by carbon nanotubes?

(Mark **ALL** correct answers)

- ☐ A) 0D
- ☐ B) 1D
- ☐ C) 2D
- ☐ D) 3D
- ☐ E) 4D

32. (3.00 pts) What is the bandgap of a semiconductor?

(Mark **ALL** correct answers)

- ☐ A) The energy required to remove an electron completely from the atom.
- ☐ B) The range of energies electrons can occupy within the valence band.
- ☐ C) The potential barrier between two semiconductor junctions.
- ☐ D) The energy difference between the valence band and the conduction band.
- ☐ E) The energy difference between donor and acceptor impurity levels.
- ☐ F) The energy required to add an electron to the atom.

33. (3.00 pts) What is the bandgap of a semiconductor?

(Mark **ALL** correct answers)

- ☐ A) $E_g = 0.001 – 0.05$ eV
- ☐ B) $E_g = 0.1 – 3.0$ eV
- ☐ C) $E_g = 3.5 – 8.0$ eV

- D) Eg = 2 – 4 meV
- E) Eg = 0.1 – 0.8 meV
- F) Eg = 0.5 – 1 MeV

34. (3.00 pts) What is the curie temperature?

(Mark **ALL** correct answers)

- A) The temperature at which water melts and boils at the same time.
- B) The critical temperature at which a ferromagnetic material loses its magnetization and becomes paramagnetic.
- C) The temperature at which a metal transitions from a normal state to a superconducting state.
- D) The critical temperature below which a semiconductor changes from insulating to conducting behavior.
- E) The temperature above which a dielectric material undergoes electrical breakdown and becomes conductive.
- F) The critical temperature above which a ferroelectric material becomes paraelectric and loses piezoelectricity.

35. (3.00 pts) What are the benefits of wet etching?

(Mark **ALL** correct answers)

- A) It's scalable, cost-effective, and suitable for batch processing.
- B) It provides highly anisotropic profiles with atomic-level precision.
- C) It offers high etch selectivity and fast etch rates for many materials.
- D) It is simple to set up and can process multiple wafers simultaneously.
- E) It eliminates the need for chemical waste handling.
- F) It requires vacuum systems and plasma sources to achieve uniform etching.

36. (3.00 pts) What are the downsides of dry etching?

(Mark **ALL** correct answers)

- A) It can cause surface damage and defects due to energetic ion bombardment.
- B) It requires complex, expensive equipment and vacuum systems.
- C) It offers extremely high selectivity between all materials under all conditions.
- D) It may produce non-uniform etch rates and sidewall residues.
- E) It can lead to contamination and redeposition of etched by-products.
- F) It eliminates the need for hazardous or reactive gases.

37. (3.00 pts)
In the sol–gel process for fabricating oxide nanomaterials, which of the following statements most accurately explains the role of hydrolysis and condensation reactions in determining the final microstructure of the gel?

(Mark **ALL** correct answers)

- A) Condensation exclusively produces volatile by-products that limit network growth, while hydrolysis primarily reduces particle size through surface oxidation.
- B) Hydrolysis and condensation govern the polymerization of metal–oxygen–metal networks; influencing the gel's homogeneity, porosity, and particle connectivity.
- C) Hydrolysis controls the degree of polymer cross-linking, while condensation determines the rate of solvent evaporation and the film's porosity.
- D) Both hydrolysis and condensation reactions occur only after the drying step, controlling densification during sintering, affecting the gel's optical transparency.

38. (3.00 pts) A type of Janus particle is

(Mark **ALL** correct answers)
- A) Hydrophobic
- B) Hydrophilic
- C) Hydrogen

39. (3.00 pts)
Ella is getting tired of material science, so she instead studies for her AP Statistics class, where they are learning about surface areas and volumes. In front of her is a cube with 6 sides, a cylinder that is round, and a circular sphere. She must pick the one that has the highest surface to volume ratio given a radius/side of length r, or else she will have to eat a raw onion. Please help her pick, she really doesn't want to eat the onion.

(Mark **ALL** correct answers)
- A) Sphere
- B) Cylinder
- C) Cube
- D) Hyperboloid

40. (3.00 pts) Which of the following factors primarily determine the final film thickness in a spin coating process?

(Mark **ALL** correct answers)
- A) Spin speed, solution viscosity, and solvent evaporation rate.
- B) Substrate temperature, grain size, and molecular weight of the solute.
- C) Chamber pressure, deposition angle, and magnetic field strength.
- D) Droplet size, electrical conductivity, and substrate reflectivity.
- E) Surface charge, plasma power, and ion bombardment time.
- F) Rainbow fish

41. (3.00 pts) Use proper grammar please, all lowercase: Buckminsterfullerene is constructed from 12 [first blank] and 20 [second blank]

42. (3.00 pts) SWNT stands for what in the context of nanomaterials? (1st blank = S, 2nd = W, 3rd plural = NT)

I know there is one word missing technically, but just do the ones in the abbreviation.

43. (3.00 pts) Fill in the blank so that is conforms to the conventions of standard English:
A ________ is a substance that lowers the surface tension of a liquid, facilitating the formation of emulsions between liquids such as oil and water.

44. (3.00 pts)
A ___-type semiconductor is formed when a trivalent impurity like boron is added to a pure semiconductor like silicon. The impurity creates “holes” in the valence band, making holes the majority charge carriers.

45. (5.00 pts) A spherical gold nanoparticle has a diameter of 10nm. Calculate its surface area to volume ratio.

46. (5.00 pts)
A CVD-grown multilayer graphene film on SiO_2 (n = 1.46 at 532 nm) is characterized by several instruments. AFM in tapping mode (spring constant 40 N/m, tip radius 8 nm, scan size 3 μm × 3 μm at 1024 lines, RMS roughness 0.18 nm) measures a step height between an oxygen-plasma–etched window and the pristine film of 1.005 nm. Raman spectroscopy (532 nm excitation, 1 mW at focus, 50× objective, spot ~0.9 μm) shows $I_2D/IG = 0.42$ and a 2D FWHM of 58 cm^{-1}. TEM cross-section imaging (200 kV, electron dose 10^3 e^-/nm^2, magnification 200k×) confirms turbostratic alignment with an interlayer spacing of 0.335 nm. The substrate oxide thickness is 285 nm, contact angle is 64°, and sheet resistance is 820 Ω/m at 300 K.

Assuming the AFM step height equals the graphene stack thickness and the interlayer spacing is uniform, how many graphene layers are present in the film?

47. (5.00 pts)
A nanowire of titanium dioxide (TiO_2) with a diameter of 80 nm and a length of 12 μm is subjected to a tensile load during nano-mechanical testing inside a TEM. The applied force increases linearly up to 24 μN, at which point the elongation measured is 18 nm. Assume the deformation is purely elastic, the cross-section is perfectly circular, and TiO_2 behaves isotropically at this scale.

Determine the Young's modulus (E) of the nanowire.

48. (5.00 pts) Find the electrical conductivity of a semiconductor with carrier concentration of $5*10^{15}$ / m^3 and electron mobility of 1350m^2/V.

49. (5.00 pts)
A solid silicon nanowire has a diameter of 60 nm and a length of 10 μm. Silicon's density is 2.33 g/cm^3. Assuming the wire is a perfectly cylindrical, fully dense solid, what is the mass of one nanowire in picograms (pg)?

50. (5.00 pts)
A 0.25kg block of material at 150ºC is placed in 0.2L of of water at 25ºC. The specific heat of the material is 0.385 J/g while the specific heat of water is 4.18 J/g. Find the final equilibrium temperature of the system.

51. (5.00 pts)
Ella meets with George Washingmachine to discuss issues with the federal bank. Alexander Hamilton admires Ella efforts in supporting the federalist party and gifts her 3 apples. Ella, excited by this gift, accidentally drop her hundred million dollar photolithography machine into the Chesapeake Bay. What is the positive difference in the number of apples sh has and the number of photolithography machines she has?

52. (5.00 pts)
My co-test writer gets carried away into writing test questions that are slightly off topic and he finds a green (λ = 550nm) photon of light exiting his computer! What is the energy of the photon in eV? (hint: 1eV = 1.602*10^-19 J)

53. (5.00 pts) Find the atomic packing density of vanadium (BCC).

54. (5.00 pts) A material sample has an original length of 0.70 m. When subjected to tension, it stretches to a maximum length of 0.75 m.
Calculate the maximum strain experienced by the material.

55. (1.00 pts) How do you ask a geologist to identify a rock?
xkcd.com/3068 (pls dont go out of browser)

- A) "Can you identify this rock that I found?"
- B) "Can you identify this rock that I found wrapped in a $5 bill?"

https://xkcd.com/2856/